TAREA M52 – CD-YAIR PALOMARES

```python
#Importación de librerías
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")

import os
os.chdir("C:/Users/yair_/OneDrive/Desktop/Ciencia de Datos/Módulo 52 - Análisis de regresión")
```

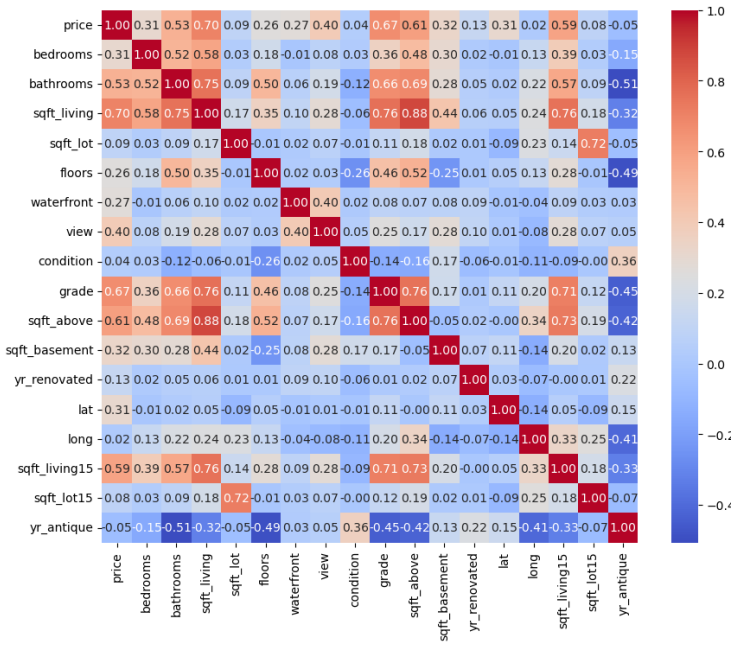
```python
#Se lee la base de datos y se convierte en dataframe
data_house = pd.read_csv("kc_house_data.csv")
data_house
```

	id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	...	grade	sqft_above	sqft_basement	yr_built	yr
0	7129300520	20141013T000000	221900.0	3	1.00	1180	5650	1.0	0	0	...	7	1180	0	1955	
1	6414100192	20141209T000000	538000.0	3	2.25	2570	7242	2.0	0	0	...	7	2170	400	1951	
2	5631500400	20150225T000000	180000.0	2	1.00	770	10000	1.0	0	0	...	6	770	0	1933	
3	2487200875	20141209T000000	604000.0	4	3.00	1960	5000	1.0	0	0	...	7	1050	910	1965	
4	1954400510	20150218T000000	510000.0	3	2.00	1680	8080	1.0	0	0	...	8	1680	0	1987	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
21608	263000018	20140521T000000	360000.0	3	2.50	1530	1131	3.0	0	0	...	8	1530	0	2009	
21609	6600060120	20150223T000000	400000.0	4	2.50	2310	5813	2.0	0	0	...	8	2310	0	2014	
21610	1523300141	20140623T000000	402101.0	2	0.75	1020	1350	2.0	0	0	...	7	1020	0	2009	
21611	291310100	20150116T000000	400000.0	3	2.50	1600	2388	2.0	0	0	...	8	1600	0	2004	
21612	1523300157	20141015T000000	325000.0	2	0.75	1020	1076	2.0	0	0	...	7	1020	0	2008	

21613 rows × 21 columns

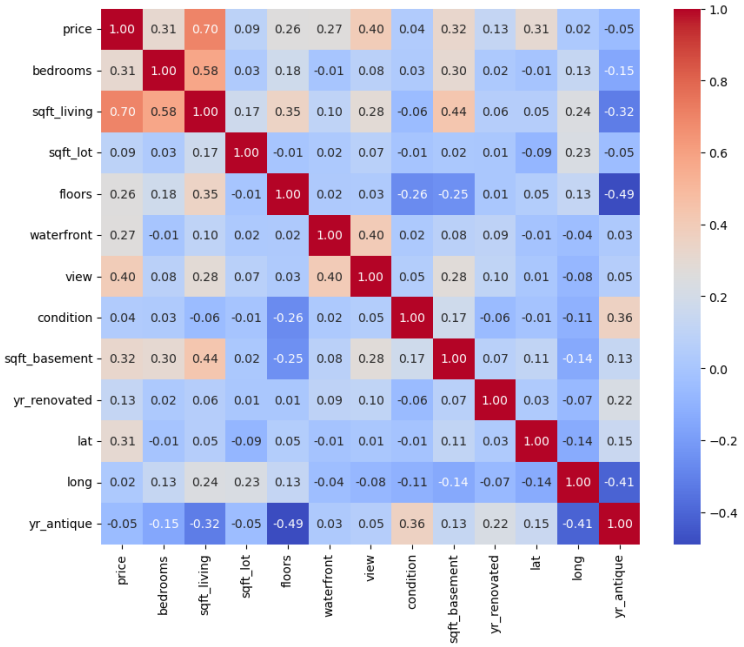
```python
#Limpieza de datos

#Se eliminan del dataframe las variables de id y zipcode. Ya que estas no son características directas con la casa
data_house = data_house.drop(columns=['id', 'zipcode'])

#Se toma el año de venta de la columna date, se convierte a int despues
data_house['date'] = data_house['date'].str[:4]
data_house['date'] = data_house['date'].astype(int)

#Se crea columna de antiguedad de la casa
data_house["yr_antique"] = data_house["date"] - data_house["yr_built"]

#Se eliminan las columnas de "date" y "yr_built"
data_house = data_house.drop(columns = ["date", "yr_built"])
```

```python
#Análisis de correlaciones entre las variables para identificar colinealidad
corr = data_house.corr(numeric_only=True)

plt.figure(figsize=(10,8))
sns.heatmap(corr, cmap="coolwarm", annot=True, fmt=".2f")
plt.show()
```



```python
#Se eliminan las variables que pueden representar problemas de colinealidad(sqft_living15, sqft_lot15, grade, bathrooms y sqft_above)
data_house = data_house.drop(columns=['sqft_living15', 'sqft_lot15', 'grade', 'sqft_above', 'bathrooms'])
```

```python
#Análisis de correlaciones entre las variables para identificar si aún existe colinealidad
corr = data_house.corr(numeric_only=True)

plt.figure(figsize=(10,8))
sns.heatmap(corr, cmap="coolwarm", annot=True, fmt=".2f")
plt.show()
```




```python
#Agregando la variable de 1 para el intercepto
data_house.insert(0, 'Intercepto', 1)
```

```python
#Definiendo las matrices
Xdata = data_house[["Intercepto", "bedrooms", "sqft_living", "sqft_lot", "floors", "waterfront",
                    "view", "condition", "sqft_basement", "yr_renovated", "yr_antique", "lat", "long"]].values
Ydata = data_house[["price"]].values
```

```python
#Dividir en bases de entrenamiento y prueba
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(Xdata, Ydata, test_size = 0.3, random_state = 1)

x = x_train
y = y_train
```

```python
#Opción de formato numérico para evitar que se coloquen decimales con notación científica
np.set_printoptions(formatter={"float_kind":"{:f}".format})
```

```python
#Reporte automatizada de la regresión en python
import statsmodels.api as sm

regressor = sm.OLS(y, x). fit()
print(regressor.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.660
Model:                            OLS   Adj. R-squared:                  0.659
Method:                 Least Squares   F-statistic:                     2441.
Date:                Sat, 14 Mar 2026   Prob (F-statistic):               0.00
Time:                        12:21:32   Log-Likelihood:            -2.0664e+05
No. Observations:               15129   AIC:                         4.133e+05
Df Residuals:                   15116   BIC:                         4.134e+05
Df Model:                          12
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const      -5.196e+07   1.81e+06    -28.741      0.000   -5.55e+07   -4.84e+07
x1         -3.835e+04   2224.506    -17.240      0.000   -4.27e+04    -3.4e+04
x2           300.7493      3.006    100.043      0.000     294.857     306.642
x3            -0.0876      0.041     -2.156      0.031      -0.167      -0.008
x4          2.591e+04   4195.535      6.175      0.000    1.77e+04    3.41e+04
x5          5.333e+05   2.12e+04     25.178      0.000    4.92e+05    5.75e+05
x6          6.473e+04   2567.456     25.213      0.000    5.97e+04    6.98e+04
x7          3.561e+04   2864.595     12.432      0.000       3e+04    4.12e+04
x8           -70.3892      5.158    -13.645      0.000     -80.500     -60.278
x9            32.9923      4.454      7.408      0.000      24.263      41.722
x10         1225.9441     80.527     15.224      0.000    1068.102    1383.787
x11         6.744e+05   1.26e+04     53.508      0.000     6.5e+05    6.99e+05
x12        -1.613e+05   1.42e+04    -11.349      0.000   -1.89e+05   -1.33e+05
==============================================================================
Omnibus:                     9385.899   Durbin-Watson:                   2.026
Prob(Omnibus):                  0.000   Jarque-Bera (JB):           291848.065
Skew:                           2.475   Prob(JB):                         0.00
Kurtosis:                      23.940   Cond. No.                     4.93e+07
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 4.93e+07. This might indicate that there are
strong multicollinearity or other numerical problems.
```

```python
#Proceso de modelo de regresión múltiple mediante matrices

#Operaciones para obtener la matriz b
XT_X = np.matmul(np.matrix.transpose(x), x)
XT_X_inv = np.linalg.inv(XT_X)

XT_Y = np.matmul(np.matrix.transpose(x), y)
betas = np.matmul(XT_X_inv, XT_Y)
betas
```

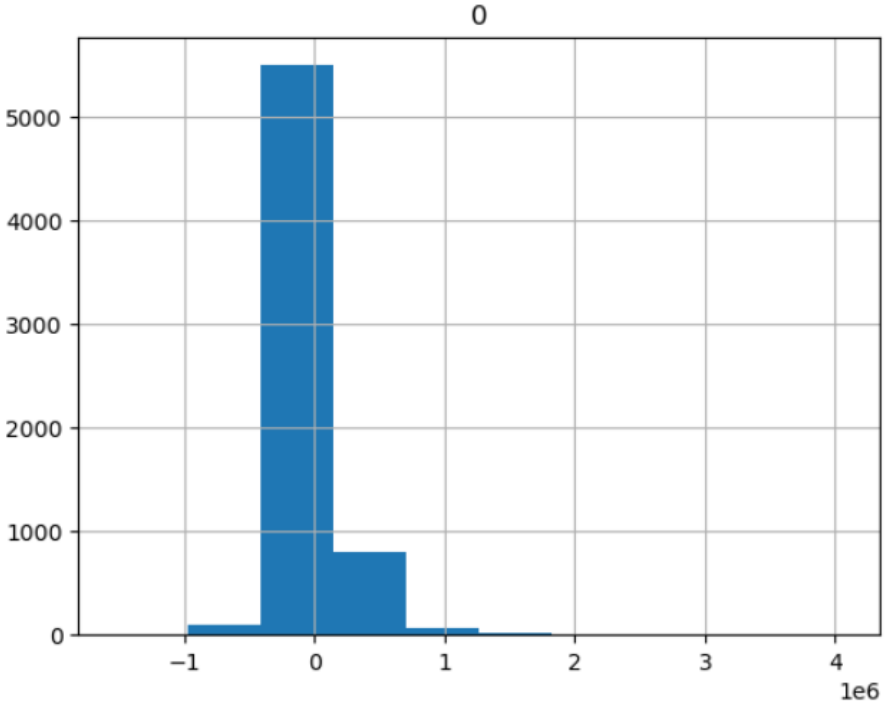
```
array([[-51956226.332031],
       [-38351.281993],
       [300.749265],
       [-0.087578],
       [25908.694079],
       [533252.631496],
       [64734.498883],
       [35613.806817],
       [-70.389190],
       [32.992305],
       [1225.944119],
       [674367.818583],
       [-161296.067377]])
```

```python
#Cálculo de TSS (Suma Total de Cuadrados)
TSS = np.matmul(np.matrix.transpose(y), y) - len(y) * (y.mean()**2)

#Cálculo del ESS (Suma Explicada de Cuadrados)
ESS = np.matmul(np.matmul(np.matrix.transpose(betas), np.matrix.transpose(x)), np.matmul(x,betas)) - len(y) * (y.mean()**2)

#Cálculo de RSS (Residuales al Cuadrado)
RSS = TSS - ESS

#Cálculo del Coeficiente de Determinación R cuadrdada
RSq = 1 - RSS / TSS
print("Coeficiente de determinación R cuadrada = ", RSq)

#Cálculo del Coeficiente de Determinación R cuadrada ajustada
RSqAj = 1 - (RSS / (x.shape[0] - x.shape[1])) / (TSS / (x.shape[0] - 1))
print("Coeficiente de determinación R cuadrada ajustada = ", RSqAj)
```

```
Coeficiente de determinación R cuadrada =  [[0.659587]]
Coeficiente de determinación R cuadrada ajustada =  [[0.659317]]
```

```python
#Aplicación sobre la base de prueba
y_pred = np.matmul(x_test, betas)

#Cálculo de residuales
Resid = y_test - y_pred

#Gráfico del histograma de residuales para la base de prueba
df = pd.DataFrame(Resid)
df.hist()
```

```
array([[<Axes: title={'center': '0'}>]], dtype=object)
```



Conclusiones finales:

De acuerdo al coeficiente de determinación R cuadrada ajustada, este modelo estadístico de regresión lineal múltiple es capaz de explicar el 65.9% del comportamiento en el precio de las viviendas.

El modelo se vería representado a través de la siguiente ecuación:

Precio estimado de la vivienda = *-51956226.33 - 38351.28*bedrooms + 300.75*sqft_living - 0.09*sqft_lot + 25908.69*floors + 533252.63*waterfront + 64734.5*view + 35613.81*condition - 70.39*sqft_basement + 32.99*yr_renovated + 1225.94*yr_antique + 674367.82*lat - 161296.07*long + error*

```python
from sklearn.metrics import r2_score
from sklearn import metrics

print("Coeficiente R cuadrado", r2_score(y_test, y_pred))
```

```
Coeficiente R cuadrado 0.6483890669904986
```